SHAKTHI MAHENDRA J

Bengaluru, India | +91 6360371224 | shakthimahendra3@gmail.com | github.com/Shakthimahendra-J

PROFESSIONAL SUMMARY

BCA graduate (2026, 8.12 CGPA) with hands-on experience in Python, Java, and web development, spanning full-stack platforms (PawCare Clinic), machine learning (Vision Transformer-based Brain Tumor Detection), AI tooling (AI Podcast Generator) and core programming (C-based Bus Reservation System). Seeking an entry-level Software/IT role to apply core programming and problem-solving skills in a production environment.

EDUCATION

Bachelor of Computer Application (BCA) / Completed, 2026
Dr. Manmohan Singh Bengaluru City University, Bengaluru, India
CGPA: 8.12

TECHNICAL SKILLS

- Programming Languages: Python, Java, C
- Web Technologies: HTML, CSS, JavaScript
- Databases & Tools: SQL, MS Excel, MS Word, Power BI
- Development Tools: VS Code, Eclipse
- Machine Learning & Data: Vision Transformers (ViT), scikit-learn, OpenCV, Matplotlib, model evaluation and tuning
- AI & LLM Tooling: LLM APIs (Minimax, OpenRouter), prompt engineering, text-to-speech (Minimax Speech 2.6), Firecrawl (web scraping), Streamlit
- Databases: MySQL (RDBMS), SQLite
- APIs & Integrations: REST APIs, JWT Authentication, Cloudinary, Nodemailer

PROJECTS

PawCare Clinic - Pet Healthcare & Pharmacy Platform | *Node.js, Express, SQLite, JavaScript, Socket.io*

- Engineered a full-stack platform (REST API design, backend, database) unifying veterinary appointments, pharmacy ordering, and pet records for customers, vets, and admins.
- Implemented JWT- and bcrypt-based authentication in Node.js with role-based access control (RBAC) across three user roles, applying secure authentication and authorization principles.
- Built a real-time chat system using Socket.io with a subscription-based access model, covering 8 REST API endpoints validated by 14/14 passing unit tests.
- Designed a normalized relational database schema (SQLite) with 6 interconnected tables and foreign key constraints, applying database design principles.
- Developed REST APIs for appointment booking, pharmacy inventory management, and order processing using Git/GitHub version control.

AI Podcast Generator | *Python, Streamlit, LLM APIs* | *In Progress*

- Engineered an automated content pipeline integrating web scraping (Firecrawl), LLM-based text generation, and text-to-speech (TTS) APIs, applying Python automation to cut manual production time by 70%.
- Designed multi-speaker prompt engineering structures, improving script reliability across 5 test articles.

Brain Tumor Detection using Machine Learning | *Python, Vision Transformer (ViT), OpenCV, Matplotlib*

- Developed a Vision Transformer (ViT) image classification model applying deep learning and computer vision techniques to detect brain tumors from MRI scan data.
- Built an OpenCV image preprocessing pipeline for noise reduction and normalization, improving data quality for model training.
- Achieved high classification accuracy; used Matplotlib to visualize performance and guide hyperparameter tuning.
- Deployed a real-time prediction pipeline for classifying new, unseen MRI scan inputs.

Bus Reservation System | *C Programming*

- Built a console-based application applying core data structures and algorithms for route search, seat selection, and ticket booking.
- Implemented seat-availability tracking and persistent data storage using structures and file handling in C.

SOFT SKILLS

- Strong communication and interpersonal skills
- Collaborative, team-first approach
- Analytical problem-solving capability

LEADERSHIP & ACTIVITIES

- Rotaract Club Member - Coordinated and managed institutional events
- Cricketer - All-rounder